

MIAE 215 Final Exam Format Overview – Programming for Engineers

Course: [Mechanical, Industrial & Aerospace engineering \(MIAE 215\)](#) 378 documents

University: [Concordia University](#)

2021/2022

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MIAE 215 Programming for Mechanical and Industrial Engineers Final Exam (all sections)

- A two-sided formula sheet and a GCS approved calculator is permitted
- **Answer ALL questions on the question pages** in the space provided
- Write your name and ID number at the top of this page
- Do not separate the question pages or the **Arduino formula sheet** (last page)

STUDENT NAME: _____

ID NUMBER: _____

Part I: Output Questions (40%)

Determine the outputs of the following programs and **write the answers in the answer boxes provided on the question pages**.

You can work outside the box or in the answer booklet. However, **only the answers inside the boxes will be marked**.

Answer any 4 of the 5 questions below. Each question is worth equal marks.

NOTE: If you answer all 5 questions we will mark only the first 4 questions.

Assume appropriate include and namespace std statements if needed.

Assume the programs have no compiler errors and they will produce output.



Part II: Programming Questions (60%)

Answer the following questions *on the question pages* in the space provided.

Include and namespace statements are not required for the programs below. However, function prototypes are required in addition to function definitions.

The Arduino question ***Q1 must be answered.*** It is worth 10%.

Then ***answer any TWO of the three questions from Q2 to Q4.*** If more than two questions are answered then ***only the first two questions will be marked.*** Each question is worth 25%.



ARDUINO FORMULA SHEET

```
void setup();  
void loop();  
  
Serial.begin(115200);  
Serial.print("\ntext");  
  
long unsigned int micros();  
long unsigned int millis();  
void delay(time_ms);  
  
Servo servo1;  
servo1.attach(pin);  
servo1.write(angle_in_deg);  
  
int analogRead(A0); // read analog pin A0 – typically returns 0 (0V) to 1023 (5V)  
  
pinMode(q,OUTPUT); // configure DIO pin q to digital output  
pinMode(q,INPUT_PULLUP); // configure DIO pin q to digital input  
  
digitalRead(q); // read digital input pin q -- returns LOW (for 0V) and HIGH (for 5.0V)  
digitalWrite(q,d); // set digital output pin q to d (LOW or HIGH)
```